

CVIS Hands-on Training on Development of Machine Learning and Deep Learning Models Summer 2025

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Paper Showcase

Paper Title: *BDConvSGNet - A Lightweight Neural Network for Bengali Digit Sound Classification*



Associated links:

Paper:

1. Main link:
<https://ieeexplore.ieee.org/document/11022527>
2. Full paper link:
https://hasan-rakibul.github.io/pdfs/islam2024bdconvsgnet.pdf?utm_source=chatgpt.com
3. Code:
<https://www.kaggle.com/code/ahsanulislam/bangla-digit-sound-classification-using-dl-kolkt>



✨ Introduction

Objective: Classify spoken Bengali digits (০-৯) using deep learning.

Why it matters:

- Bengali is the 7th most spoken language globally.
- Voice-based interfaces are essential for accessibility and smart applications in low-resource environments.
- Existing models are often heavy and unsuitable for deployment on edge devices.

BDConvSGNet: A compact deep neural network with only ~97k parameters, achieving >99% accuracy.



Problem Definition & Dataset

Task:

Identify spoken Bengali digits from audio recordings.

Dataset Overview:

Dataset	No. of Samples	Accuracy (<u>BDConvSGNet</u>)
Dataset-1	984	~94.00%
Dataset-2	2,252	~98.76%
Dataset-3	13,554	~99.20%



Preprocessing:

- Sampling rate normalized (e.g., 16kHz)
- Silence trimming
- Padding/clipping
- Feature extraction: **MFCC (Mel-Frequency Cepstral Coefficients)**

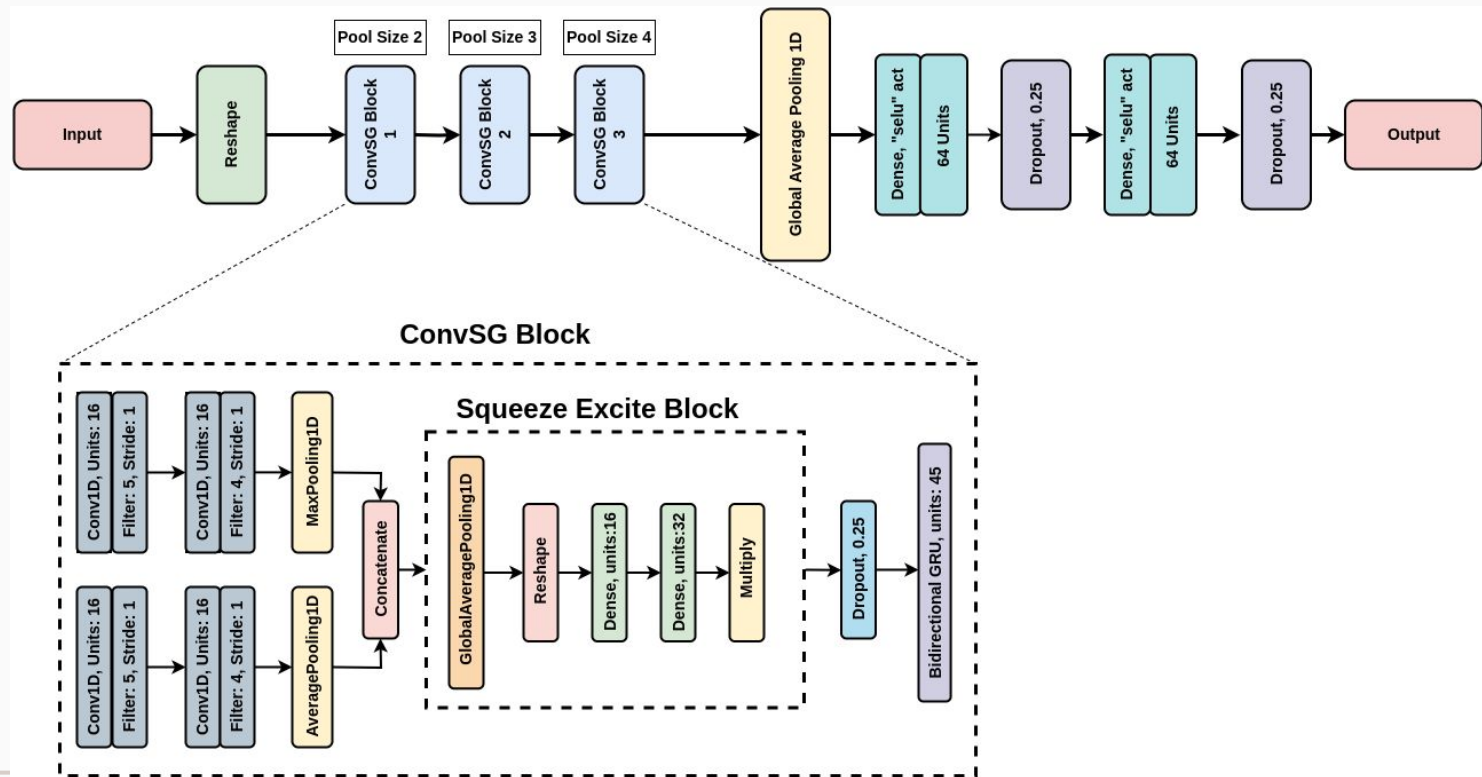
MFCC Features: Capture frequency characteristics of the human voice.

```
import librosa
import numpy as np

def extract_mfcc(filepath):
    y, sr = librosa.load(filepath, sr=16000)
    mfcc = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=40)
    return mfcc.T
```



BDConvSGNet Architecture





Training and Evaluation

Loss Function

- Categorical Cross Entropy

Optimizer

- Adam

Metrics Used

- Accuracy
- F1-score
- Cohen's Kappa
- AUC-ROC

Cross Validation

- 10-fold CV
- Low variance: proves consistency and generalizability



Performance Analysis

Model	Accuracy	Parameters
CNN + MFCC	96.7%	180k+
CNN + Spectrogram	98.23%	~200k
Hybrid Features	98.52%	?
BDConvSGNet	99.20%	97k



Deployment & Applications

Edge Deployment

- Can be exported to TensorFlow Lite / ONNX
- Low memory footprint
- Real-time classification

Use Cases

- Smart voice interfaces
- Bengali voice-controlled systems
- Accessibility tech for visually impaired



Summary

- BDConvSGNet achieves state-of-the-art accuracy with minimal resources
- Efficient architecture: CNN + Bi-GRU + SE
- Outperforms heavier models
- Scalable and deployable for real-world Bengali voice tech



Q&A